DSCC Lab midterm

Gouthaman G
CB.SC.U4CYS23014

Tree structure

```
┌──(goutham24㉿gouthaman)-[~/Clg/DSCC/CB.SC.U4CYS23014_midterm]
└─$ tree
.
├── database
│   ├── abc.sh
│   └── init.sql
├── docker-compose.yml
├── Dockerfile
├── README.md
└── web
    └── app.py

3 directories, 6 files
```

1. Dockerfile

```
FROM python:3.12-slim

ENV ROLLNUMBER=CB.SC.U4CYS23014
ENV NAME=Gouthaman

LABEL EMAIL="cb.sc.u4cys23014@cb.students.amrita.edu"
LABEL PRODUCTION_VERSION="1.0"

WORKDIR /midterm-exam

RUN apt-get update && \
   apt-get install -y mariadb-server mariadb-client && \
   pip install flask mysql-connector-python && \
   rm -rf /var/lib/apt/lists/*

COPY web /midterm-exam/web
COPY database /midterm-exam/database

RUN chmod +x /midterm-exam/database/abc.sh

EXPOSE 5000
CMD ["/midterm-exam/database/abc.sh"]
```

2. docker-compose.yml

```
services:

 webauth:
   build: .
   image: mt-webauth
   container_name: mt-webauth-container

   ports:
    - "8014:5000"

   volumes:
    - ./web:/midterm-exam/web
```

3. database/abc.sh

```
#!/bin/bash

mysqld_safe --datadir=/var/lib/mysql &

sleep 5

mysql -u root < /midterm-exam/database/init.sql

python /midterm-exam/web/app.py
```

4. database/Init.sql

```
CREATE DATABASE IF NOT EXISTS midterm_db;

CREATE USER IF NOT EXISTS 'midterm_user'@'localhost'
IDENTIFIED BY 'midterm_pass';

GRANT ALL PRIVILEGES ON midterm_db.*
TO 'midterm_user'@'localhost';

USE midterm_db;

CREATE TABLE IF NOT EXISTS users (
   id INT PRIMARY KEY AUTO_INCREMENT,
   username VARCHAR(50),
   password VARCHAR(50)
);

INSERT INTO users (username, password)
VALUES ('Goutham', 'password123');
```

5. web/app.py

```
from flask import Flask, request
import mysql.connector

app = Flask(__name__)


@app.route("/", methods=["GET", "POST"])
def login():

   message = ""

   if request.method == "POST":

       username = request.form["username"]
       password = request.form["password"]
       db =
mysql.connector.connect(host="127.0.0.1",user="midterm_user",password="
midterm_pass",database="midterm_db")
```

```
        cursor = db.cursor()
        cursor.execute("SELECT * FROM users WHERE username=%s AND
password=%s",(username, password))
        result = cursor.fetchone()
        cursor.close()
        db.close()
        if result:
            message = "Login successful!"
        else:
            message = "Invalid username or password!"

    return f"""

    <html>
    <head>
        <title>Web Authentication</title>
    </head>

    <body>

        <h1>DSCC lab midterm</h1>

        <form method="POST">

            Username:
            <input type="text" name="username">

            <br><br>

            Password:
            <input type="password" name="password">

            <br><br>

            <input type="submit" value="Login">

        </form>

        <h3>{message}</h3>

    </body>
    </html>
    """
```

```
app.run(host="0.0.0.0", port=5000)
```

6. README.md

```
login details

login- Goutham
password- password123

---------------------------------

docker commands used

1. docker build -t mt-webauth . -> used to build the image
2. docker compose up -d  -> used to compose the image
3. docker ps -a  -> used to verify whether the image is running
4. docker tag mt-webauth:latest goutham2465/mt-webauth:latest  -> used to tag the image
5. docker push goutham2465/mt-webauth:latest -> used to push the image
```

Output:

1. Building the image using the command:

Docker build -t mt-webauth .



2. Composing the image
Docker compose up -d

```
┌──(goutham24㉿gouthaman)-[~/Clg/DSCC/CB.SC.U4CYS23014_midterm]
└─$ docker compose up -d
[+] up 2/2
 ✔ Network cbscu4cys23014_midterm_default Created
 ✔ Container mt-webauth-container         Started
```

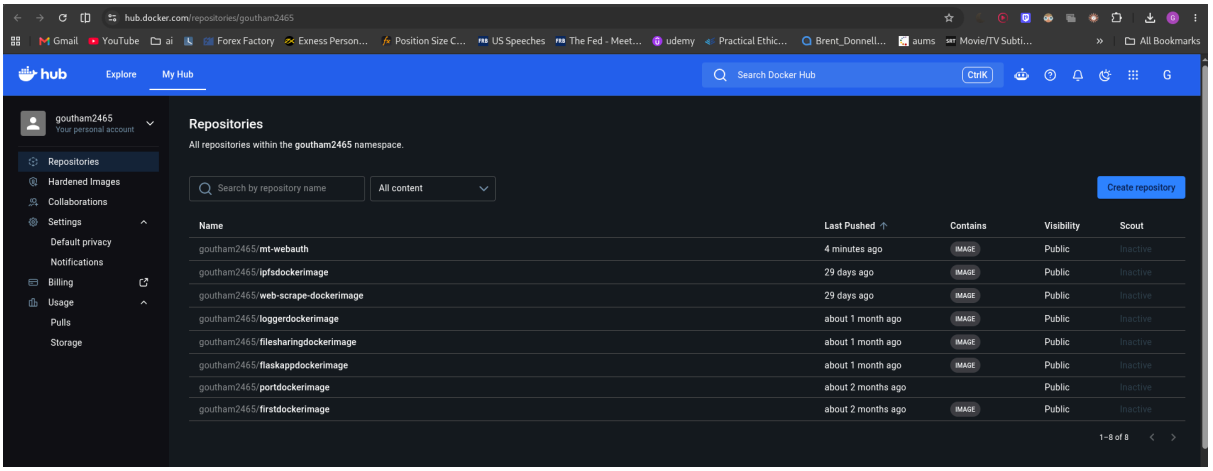
Image is running

```
┌──(goutham24㉿gouthaman)-[~/Clg/DSCC/CB.SC.U4CYS23014_midterm]
└─$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED          STATUS                          PORTS                                       NAMES
8c37bc4dd864   mt-webauth                           "/midterm-exam/datab…"   8 minutes ago    Up 8 minutes                    0.0.0.0:8014->5000/tcp, [::]:8014->5000/tcp mt-webauth-container
faaff44d5574   goutham2465/flaskappdockerimage      "python app.py"          2 weeks ago      Exited (255) About an hour ago  5000/tcp, 0.0.0.0:8017->80/tcp              flaskapp
449e506da80a   goutham2465/portdockerimage          "/docker-entrypoint.…"   2 weeks ago      Exited (255) About an hour ago  0.0.0.0:8016->80/tcp                        containerr
ad293974c8b6   goutham2465/web-scrape-dockerimage   "python scraper.py"      2 weeks ago      Exited (0) 2 weeks ago                                                      container
72bbf9658c2e   portimage                            "/docker-entrypoint.…"   6 weeks ago      Exited (0) 6 weeks ago                                                      santhosh
```

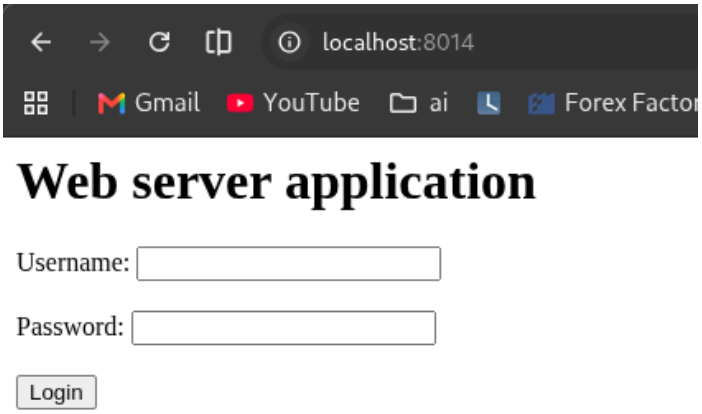
3. Pushing the repo to docker hub

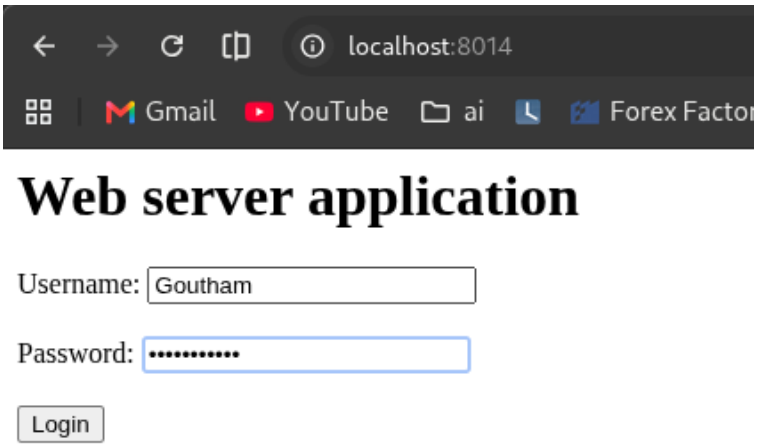
```
┌──(goutham24㉿gouthaman)-[~/Clg/DSCC/CB.SC.U4CYS23014_midterm]
└─$ docker push goutham2465/mt-webauth
Using default tag: latest
The push refers to repository [docker.io/goutham2465/mt-webauth]
fa18dfb1257a: Pushed
efa70d1e2516: Pushed
4a3b176d31be: Pushed
2c2bd0416f43: Pushed
b81384ca3614: Pushed
29fd1a28dcaf: Pushed
cbe3275cc4c5: Pushed
26c307b5e35a: Pushed
b952fe9f6810: Pushed
6760bfe2ff00: Pushed
latest: digest: sha256:f1e9897c7af72c0dbad5d1cffb4b697522e078c01212032d09ba57c08be22bc2 size: 856
```

The image has been pushed to docker hub

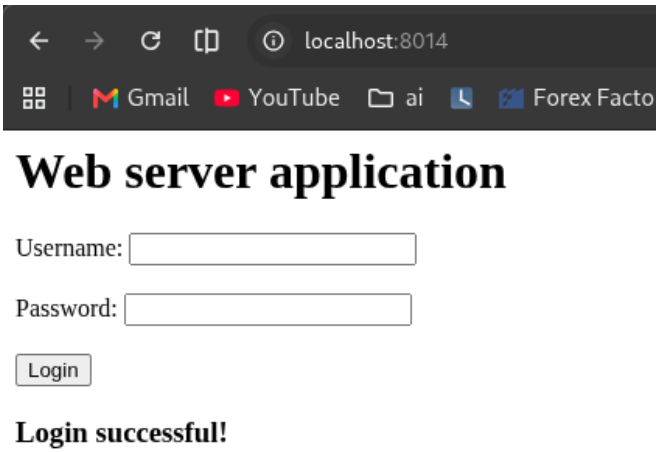


4. Opening http://localhost:8014

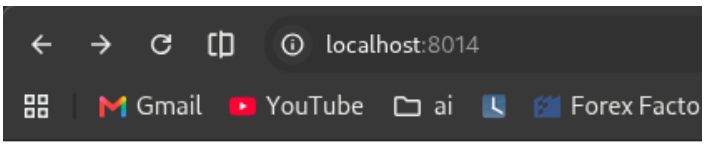




Login successful



Incorrect credentials

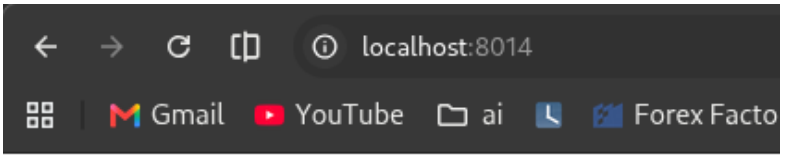


Web server application

Username: Gouth

Password: ••••

Login



Web server application

Username:

Password:

Login

Invalid username or password!

Invalid credentials

5. ENV variables

```
ENV ROLLNUMBER=CB.SC.U4CYS23014
ENV NAME=Gouthaman
```

6. LABEL

```
LABEL EMAIL="cb.sc.u4cys23014@cb.students.amrita.edu"
LABEL PRODUCTION_VERSION="1.0"
```